COMP 3920 Database Systems 1

MySQL
Party Database

Introduction:

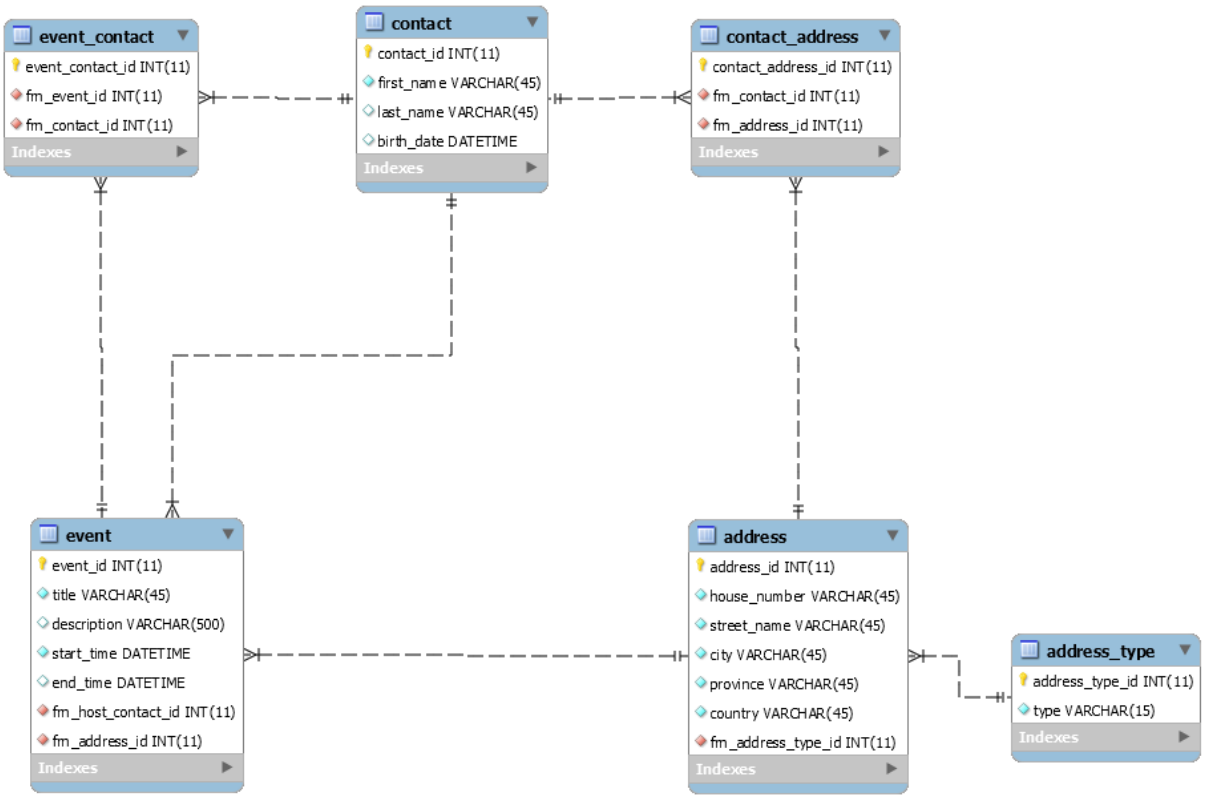
The goal of this document is to serve as a walkthrough creating a simple database with tables, constraints, foreign keys and various data types in MySQL 8.

For this document, we will be using command prompt commands and MySQL Workbench in MySQL 8.

We will be creating 6 tables:

1. contact (i.e. people)
2. address (i.e. people's houses or party venues)
3. contact_address (i.e. where do people live, or work?)
4. address_type (ex: home, work, public)
5. event (i.e. parties)
6. event_contact (i.e. Invitees - who is coming to the party?)

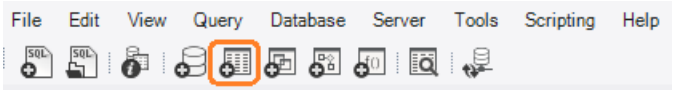
Database Entity Relationship Diagram (ERD) shown below:



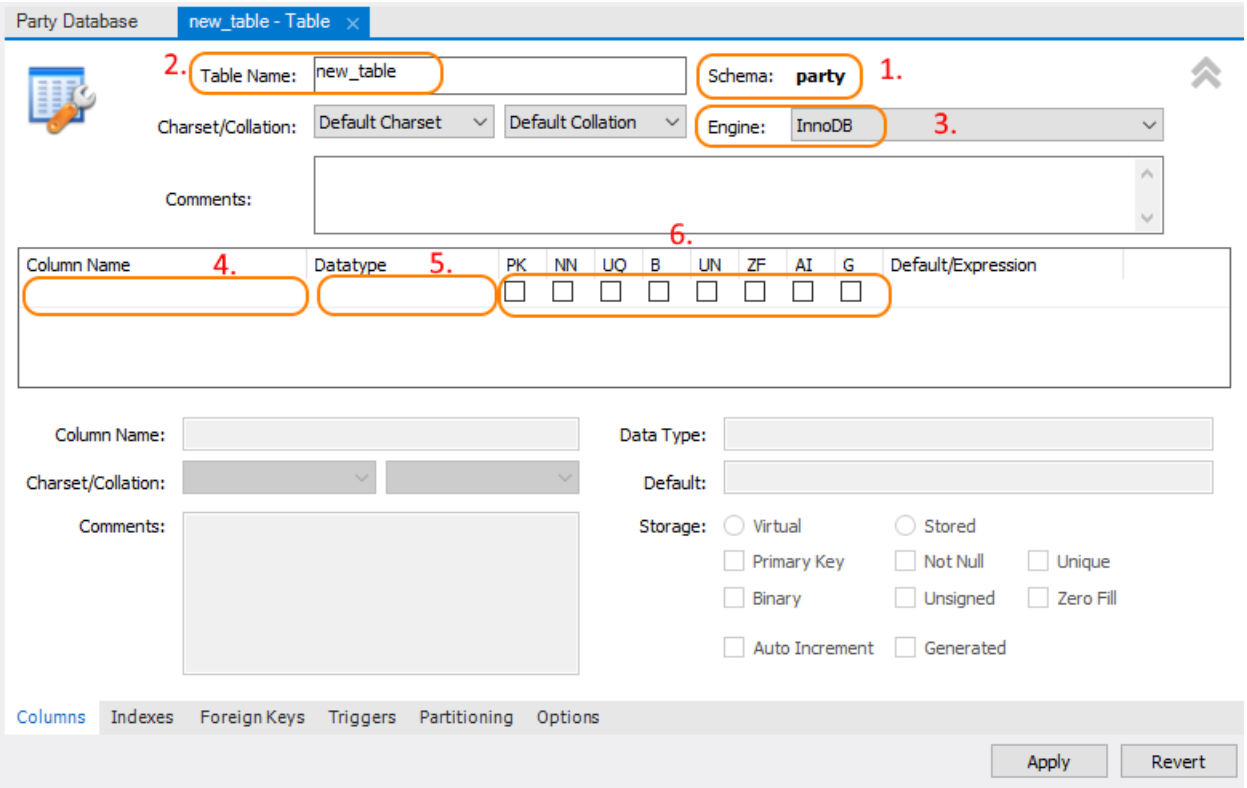
Step 1:
Open the MySQL 8 Workbench.

Step 2:
Create a new database called "party".
Set "party" as your default database.

Step 3:
From within MySQL 8 Workbench open the new table tool.



The new table tool will open as a new tab:



1. Ensure the default database is `party`.
2. Enter the name of your new table.
3. Ensure that the table storage engine is `InnoDB`.
4. For each of the columns in your table add a column name,
5. an appropriate datatype
6. and verify the correct flags are set:
    a. PK – Primary Key
    b. NN – Not Null
    c. UQ – Unique (no duplicates)
    d. B – Binary
    e. UN – Unsigned (positive numbers only)
    f. ZF – Zero Filled (show numbers as 0001, 0025, 0314, etc.)
    g. AI – Autoincrement (will use this for our primary key)
    h. G – Generated

Create the following tables:

address_type

Field	Type	Null	Key	Extra
address_type_id	int	NO	PRI	auto_increment
type	varchar(15)	NO		

address

Field	Type	Null	Key	Extra
address_id	int	NO	PRI	auto_increment
house_number	varchar(45)	NO		
street_name	varchar(45)	NO		
city	varchar(45)	NO		
province	varchar(45)	NO		
country	varchar(45)	NO		
frn_address_type_id	int	NO		

contact_address

Field	Type	Null	Key	Extra
contact_address_id	int	NO	PRI	auto_increment
frn_contact_id	int	NO		
frn_address_id	int	NO		

contact

Field	Type	Null	Key	Extra
contact_id	int	NO	PRI	auto_increment
first_name	varchar(45)	NO		
last_name	varchar(45)	YES		
birth_date	datetime	YES		

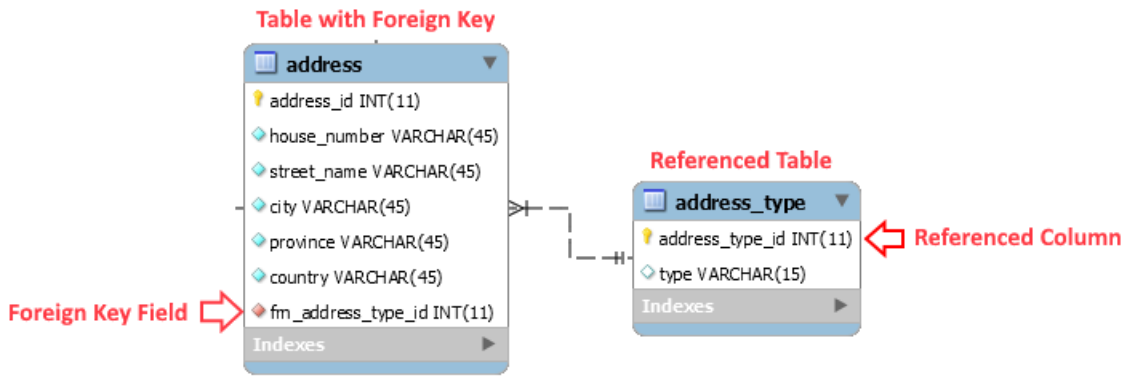
event_contact

Field	Type	Null	Key	Extra
event_contact_id	int	NO	PRI	auto_increment
frn_event_id	int	NO		
frn_contact_id	int	NO		

event

Field	Type	Null	Key	Extra
event_id	int	NO	PRI	auto_increment
title	varchar(45)	NO		
description	varchar(500)	YES		
start_time	datetime	NO		
end_time	datetime	YES		
frn_host_contact_id	int	NO		
frn_address_id	int	NO		

Step 4:
Before saving each table we will create the foreign keys as shown in the Database ERD.
Let's go through and example of creating the foreign key between `address` and `address_type`.



The `address` table has a foreign key called `frn_address_type_id`. We will be prefixing all foreign key fields with `frn`.

To create a foreign key we will follow these steps:

1. Give your Foreign Key a name – use the Table With Foreign Key Name followed by an underscore (_) and the Foreign Key Field Name:

   table name: `address`
   foreign key field name: `frn_address_type_id`
   foreign key name: `address_frn_address_type_id`

2. Select the Referenced Table:

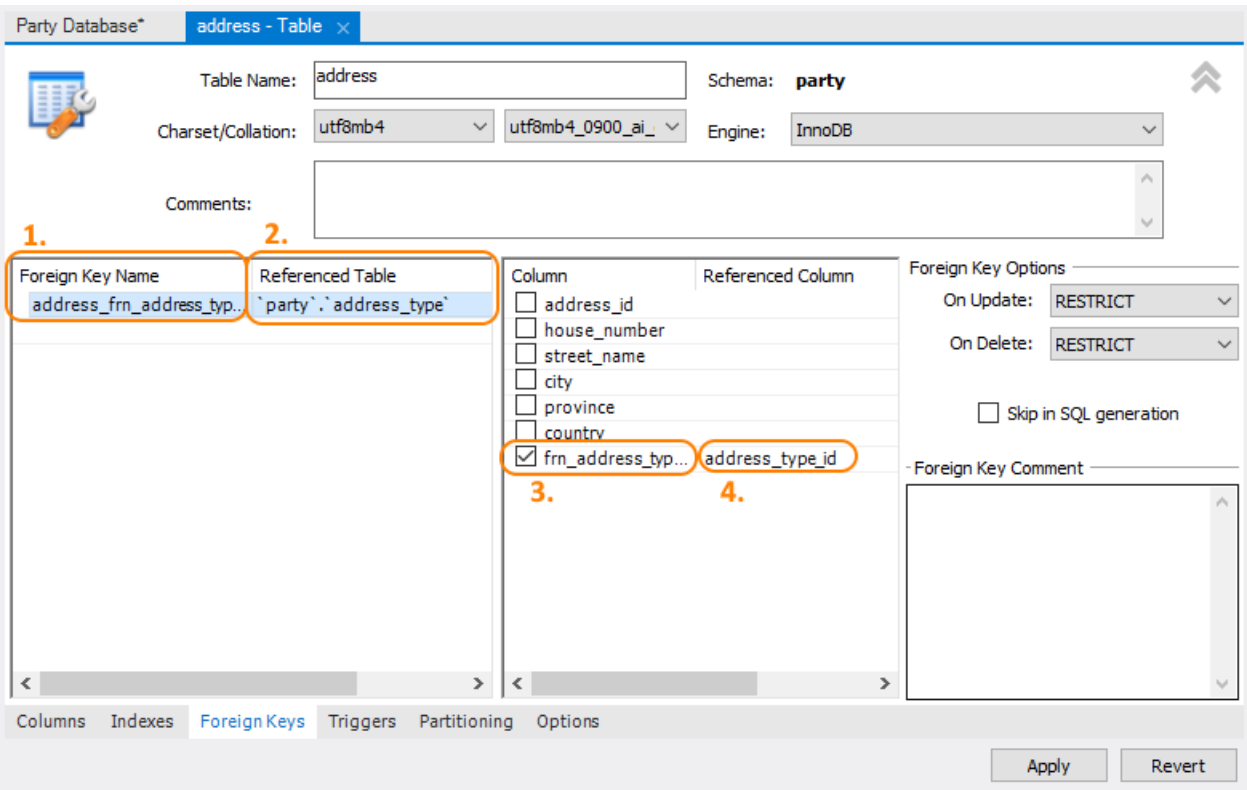
   referenced table: `address_type`

3. Select the Foreign Key Field Name:

   foreign key field name: `frn_address_type_id`

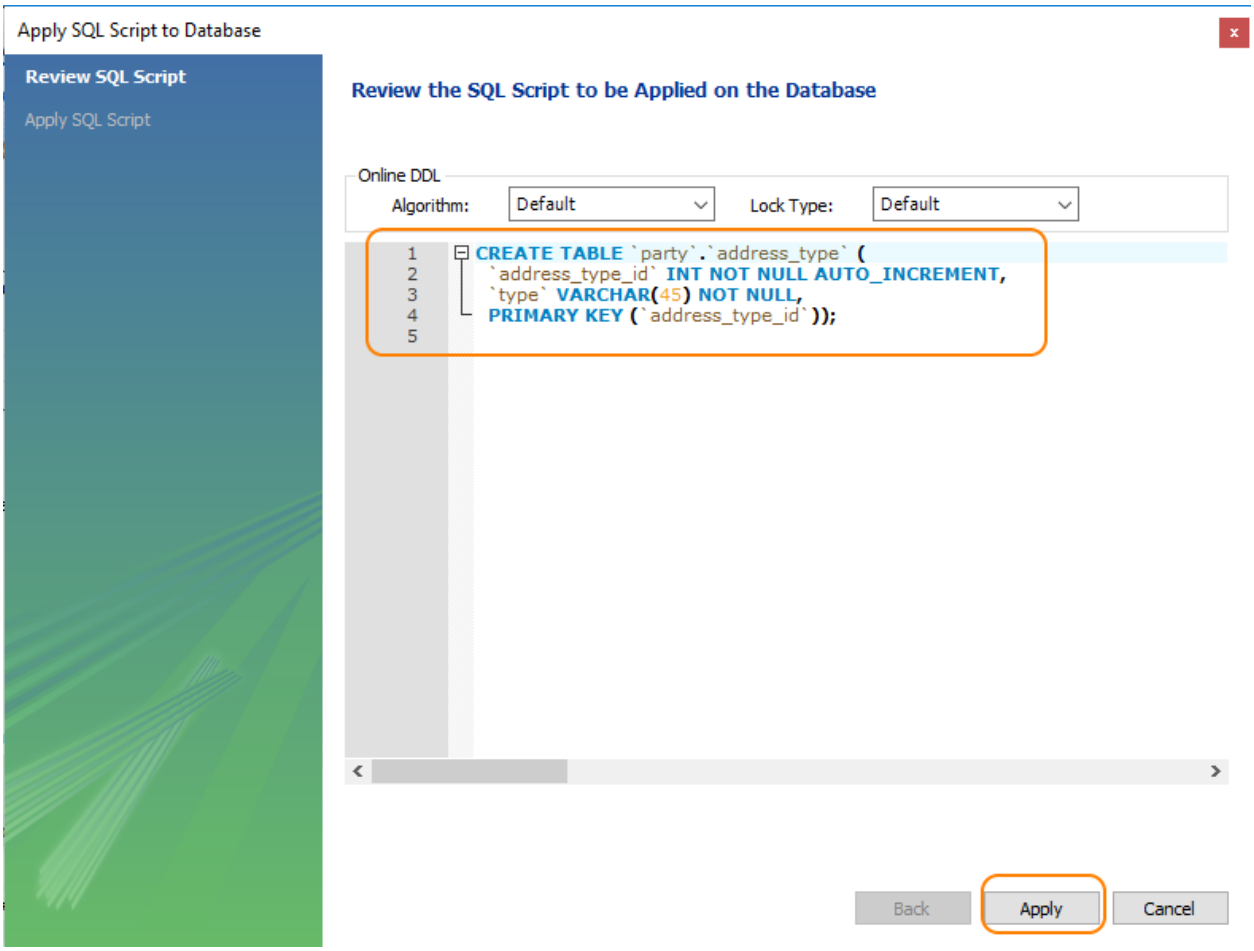
4. Select Referenced Column:

   referenced column: `address_type_id`



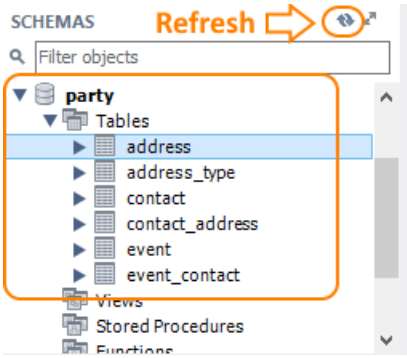
Once you have added all the columns and foreign key definitions to the new table are about to create, click Apply.

On the next screen you will see the `CREATE TABLE` statement that MySQL 8 Workbench has created for you and will attempt to run for you. Have a quick look at the statement and review it. You can edit the SQL statement directly here, or you can click Cancel to go back to the new table tool to make changes, if you see any errors or omissions. Click Apply if you are ready to proceed.



Repeat until all 6 tables have been created.

Verify the tables have been created in the 'Schema View':
Click Refresh if needed. Ensure all the tables have been added as expected.



Step 5:
Insert data into the tables. We are going to create some data to populate these tables (data shown below).

For autoincrement columns, leave the field blank.

IMPORTANT: The order we put the data into the database matters!

Why? Let's think about this for a moment. Can we invite people (add entries to the `event_contact` table) to a party that doesn't exist (empty `event` table) or before we know who the people are (empty `contact` table)? No. And the database will enforce this. Every invitee (`event_contact` entry) needs to know its event and contact information (because of the foreign key constraints).

So where do we start? A simple rule is, always start with tables that don't have a foreign key field (look for the prefix `frn`).

A good candidate is `address_type`.

Once `address_type` has some entries, we can do the `address` table (since it only has 1 foreign key and we have entries in the `address_type` table.

Add the following data to your tables:

address_type

address_type_id	type
1	work
2	home
3	public

address

address_id	house_number	street_name	city	province	country	frn_address_type_id
1	741	Heather St	Richmond	BC	Canada	2
2	10223	Shell St	Richmond	BC	Canada	2
3	#105 887	River Rd	Delta	BC	Canada	2
4	6088	Main St	Burnaby	BC	Canada	3

event

event_id	title	description	start_time	end_time	frn_host_contact_id	frn_address_id
1	Spring Break Kickoff	Pool party at the Tortia's	2018-03-15 18:00	2018-03-15 23:50	2	2

contact

contact_id	first_name	last_name	birth_date
1	Joe	Sorenson	1997-05-22 0:00
2	Sara	Tortia	1996-01-10 0:00
3	Lia	Tortia	1994-08-09 0:00
4	Michael	Park	1995-06-23 0:00

event_contact

event_contact_id	frn_event_id	frn_contact_id
1	1	1
2	1	2
3	1	3
4	1	4

contact_address

contact_address_id	frn_contact_id	frn_address_id
1	1	1
2	2	2
3	3	2
4	4	3

Run the following queries to test your tables and data:

1. ```
   SELECT
         e.title,
         ea.house_number AS "event_house_number",
         ea.street_name AS "event_street_name",
         ea.city AS "event_city",
         h.first_name AS "host_name",
         i.first_name AS "guest_name"
   FROM event AS e
   INNER JOIN address AS ea
     ON e.frn_address_id = ea.address_id
   INNER JOIN event_contact AS ec
     ON e.event_id = ec.frn_event_id
   INNER JOIN contact AS i
     ON ec.frn_contact_id = i.contact_id
   INNER JOIN contact AS h
     ON h.contact_id = e.frn_host_contact_id;
   ```
2. ```
   SELECT
         c.first_name,
         c.last_name,
         a.house_number,
         a.street_name,
         a.city
   FROM contact AS c
   INNER JOIN contact_address AS ca
     ON c.contact_id = ca.frn_contact_id
   INNER JOIN address AS a
     ON a.address_id = ca.frn_address_id;
   ```

Run the following command:

```
mysql -u root -p -t < "Show Events MySQL.sql" >
"Event_Results_MySQL.txt"
```

IMPORTANT! Make sure you include the **`-t`** option when running the above command.

You will be marked on your accuracy and attention to detail for the table structure and data.

Make sure that:

- your tables are named correctly
- you have the correct columns with the correct data types
- the primary keys in all your tables are set to `auto_increment`
- that you have all the correct foreign key constraints on your table definitions
- all the correct people are invited to the correct parties at the correct addresses
- columns in the tables are marked as NOT NULL as described in the instructions

Submit your results file:
Your file should be around **27 Kbytes**.
`01_Event_Results_MySQL.txt`

The initial contents of `Event_Results_MySQL.txt` should look like this:

```
+-----------------+
| Tables_in_party |
+-----------------+
| address         |
```

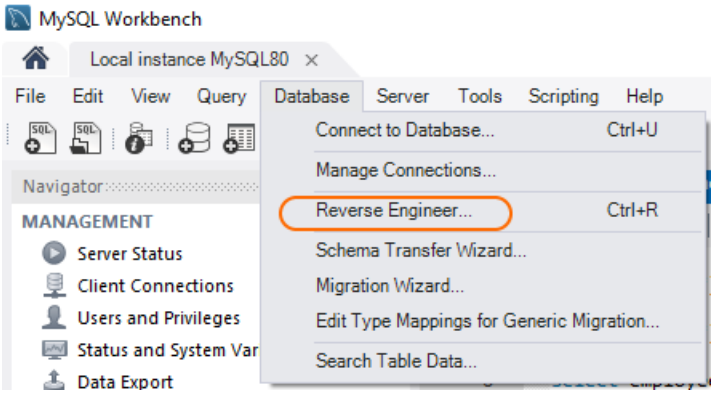
… and it should end with:

```
+------------+-----------+--------------+-------------+----------+
| first_name | last_name | house_number | street_name | city     |
+------------+-----------+--------------+-------------+----------+
| Joe        | Sorenson  | 741          | Heather St  | Richmond |
| Sara       | Tortia    | 10223        | Shell St    | Richmond |
| Lia        | Tortia    | 10223        | Shell St    | Richmond |
| Michael    | Park      | #105 887     | River Rd    | Delta    |
+------------+-----------+--------------+-------------+----------+
```

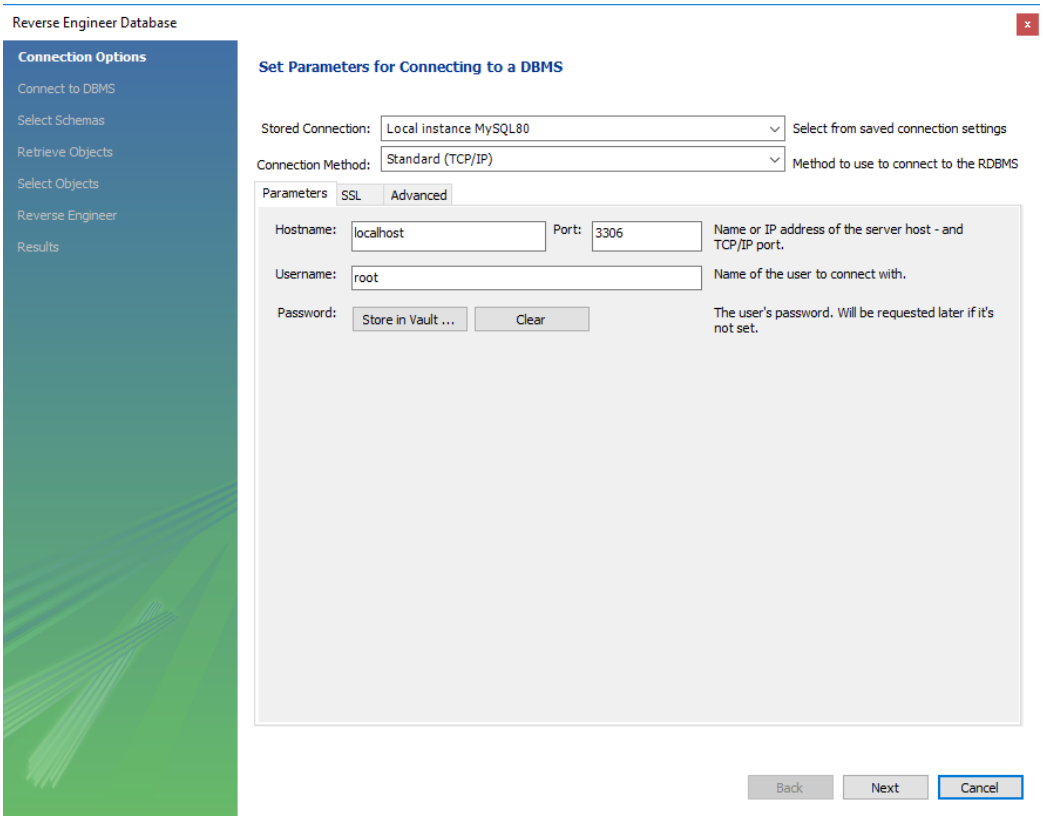
Step 6: Reverse Engineer our new database.

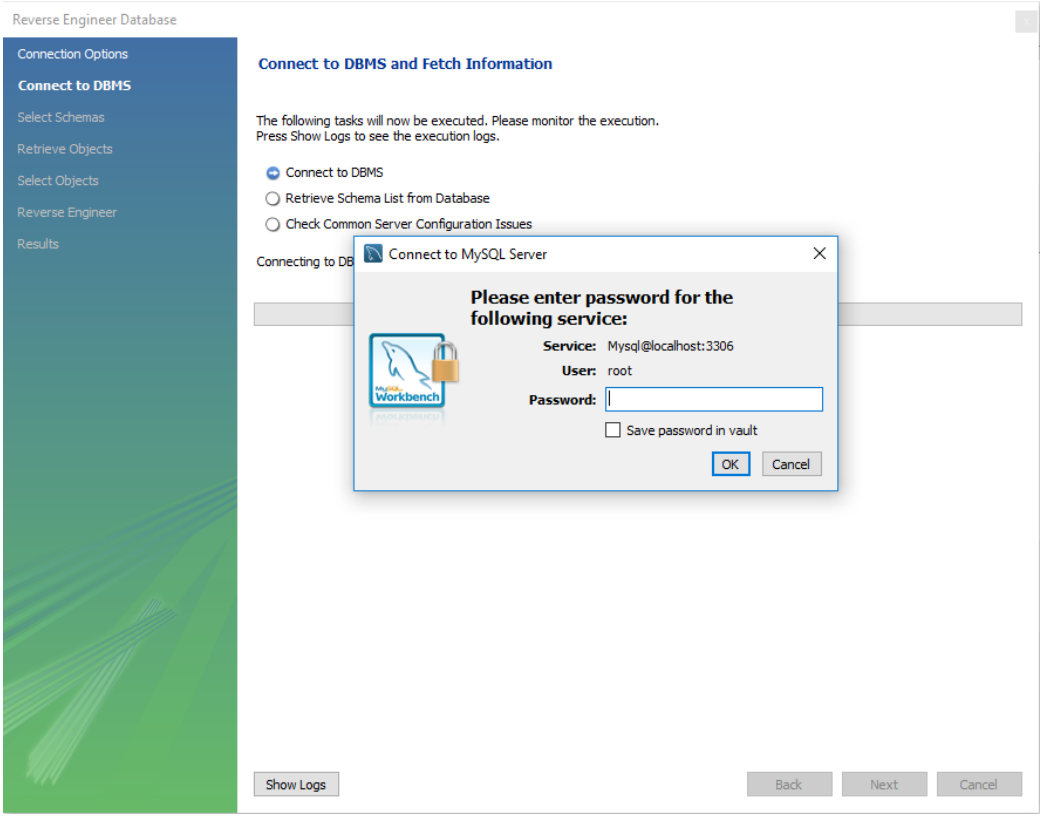
For existing databases, we can generate Entity Relational Diagrams (ERD) right from within MySQL Workbench. ERDs are useful to graphically show all the tables and foreign keys in your database.

To start click Database → Reverse Engineer…

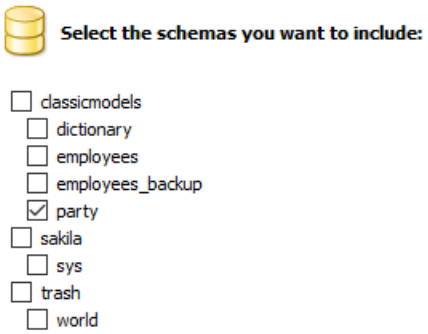


Click Next and Enter your root password

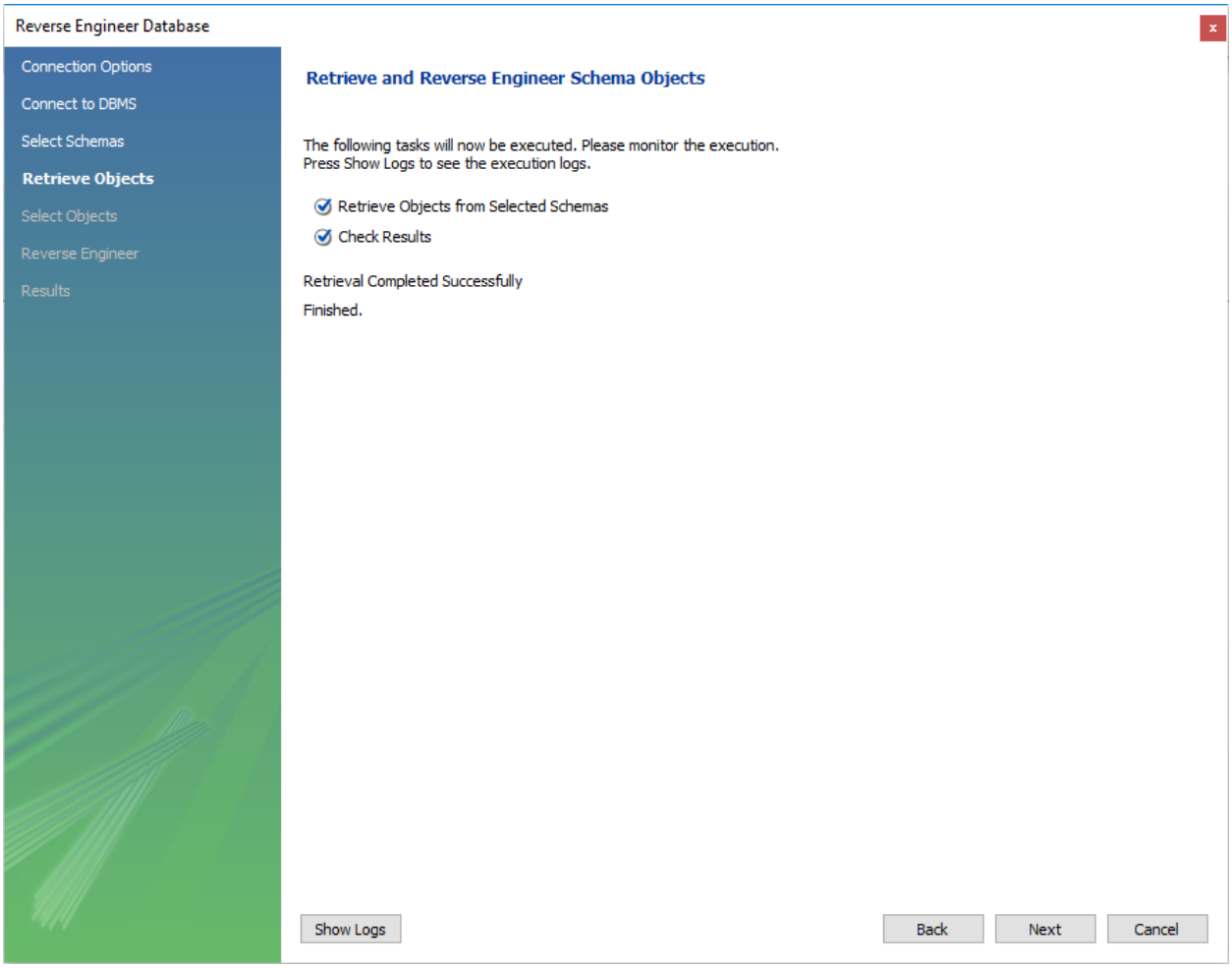




Select your database:

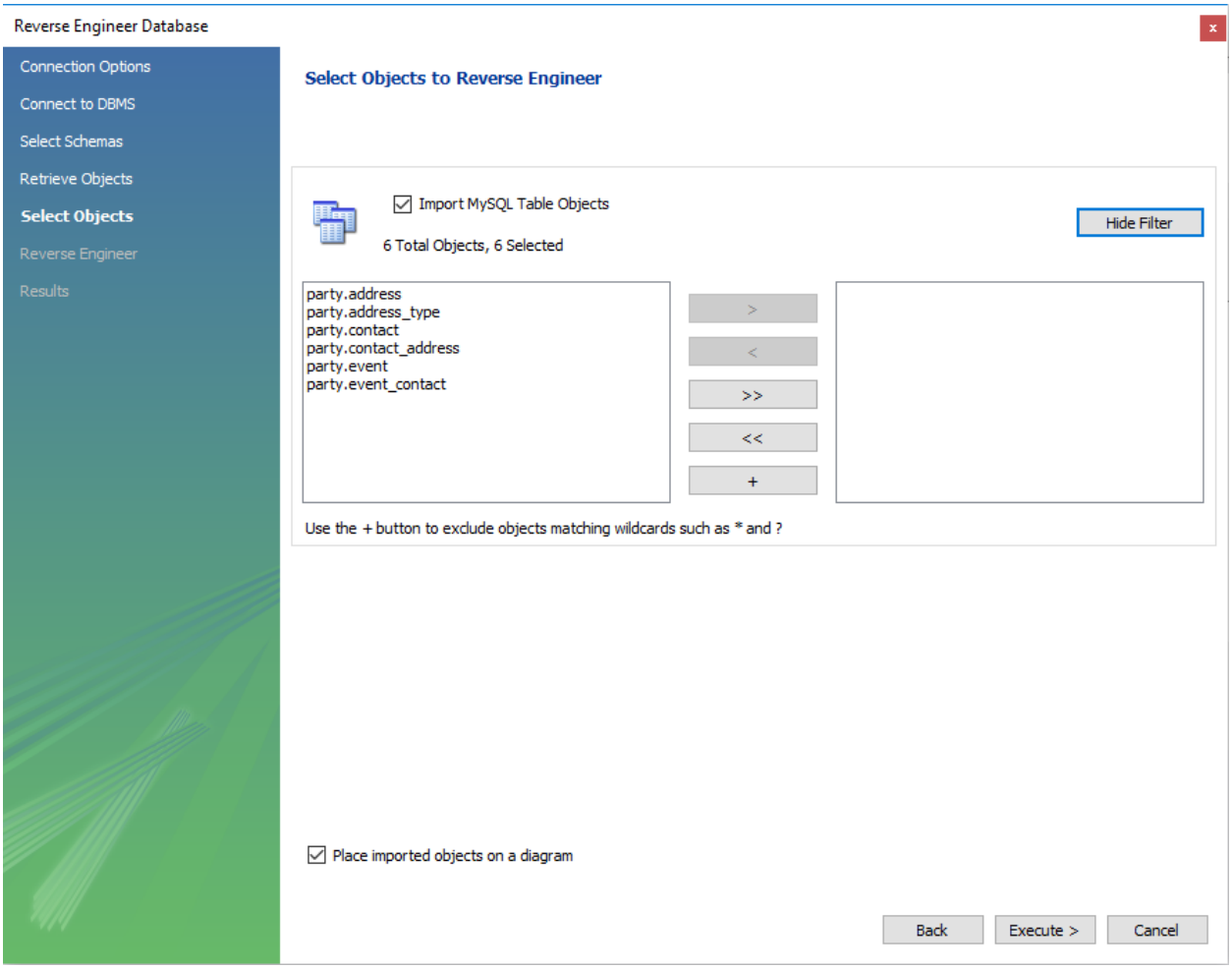


Click Next



Click Next

If you only want to show some of the tables in your diagram you can filter out some of the unwanted tables here:

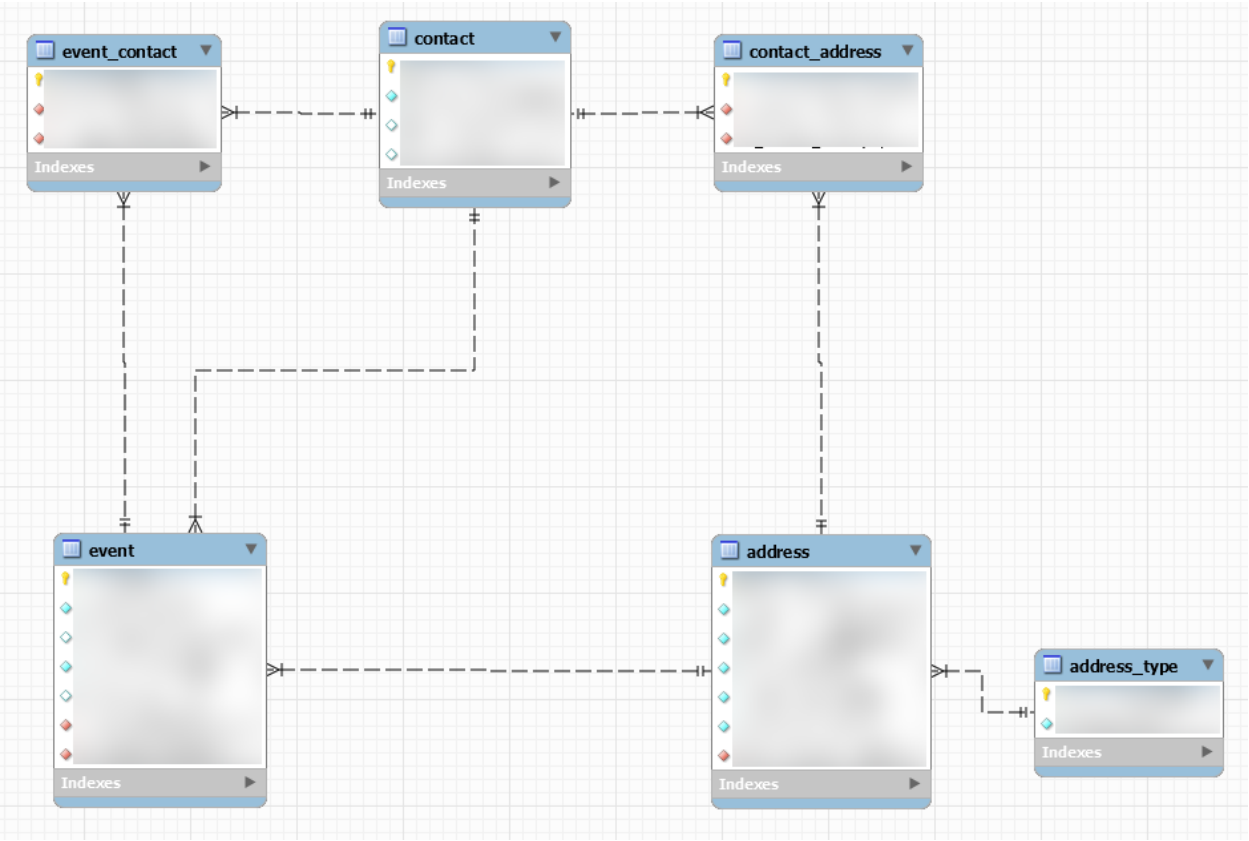


For us we want all of our new tables (ensure that they are all selected).
Click Execute
Click Next and Finish

You should now see your ERD that MySQL Workbench has created for you.

Take a moment to rearrange the tables so that the foreign key lines are not crossing each other and to clean up the diagram.

Make sure that none of the relationship lines are hidden behind other entities.

Submit your screenshot:
Filename: **02_MySQL_Reverse_Engineer.jpg**

Submit